

证明. ∇f 与等值线正交.

[证]. 设 $f = f(x, y)$. f 的等值线为 $r(t) = g(t)\vec{i} + h(t)\vec{j}$

则 f 的 C 等值线为 $f(g(t), h(t)) = C$.

$\frac{d}{dt} f(g(t), h(t)) = \frac{d}{dt} C = 0 \Rightarrow \frac{df}{dx} \frac{dx}{dt} + \frac{df}{dy} \frac{dy}{dt} = 0$

$\Rightarrow (f_x, f_y) \cdot (\frac{dx}{dt}, \frac{dy}{dt}) = 0 \Rightarrow \nabla f \cdot \vec{r}' = 0$

$\Rightarrow \nabla f$ 与等值线的切线正交 $\Rightarrow \nabla f$ 与等值线正交.

[The Derivative along a Path]. $\frac{d}{dt} f(r(t)) = \nabla f(r(t)) \cdot r'(t)$

[线性化估计]. f 的 1.2 阶偏导连续, 则 f 在 (x_0, y_0) 的 Linearization: $L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

误差为 $E(x, y) = f(x, y) - L(x, y)$, 满足 $|E(x, y)| \leq \frac{1}{2} M \cdot ((x - x_0)^2 + (y - y_0)^2)$, 其中 $M = \max\{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$

[First Derivative Test]. 对每点且 $\nabla f = 0 \Rightarrow$ 关键点. critical point. critical point $= \nabla f = 0$ 或 f_x, f_y (其中一) DNE

[Second Derivative Test]. f 的 1.2 阶偏导连续.

① $f_{xx} < 0, f_{xx}f_{yy} - f_{xy}^2 > 0 \Rightarrow$ maximum.

② $f_{xx} > 0, f_{xx}f_{yy} - f_{xy}^2 > 0 \Rightarrow$ minimum.

③ $f_{xx}f_{yy} - f_{xy}^2 < 0 \Rightarrow$ saddle point.

④ $f_{xx}f_{yy} - f_{xy}^2 = 0 \Rightarrow$ unknown.

其中 $f_{xx}f_{yy} - f_{xy}^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = |Hf|$

有界闭上的最大/小值: 在边界点或关键点上.

Taylor Formula.

设 f 有 $n+1$ 阶连续偏导, 则 f 在 (x_0, y_0) 的 Taylor 展开:

$f(a+h, b+k) = f(a, b) + (h f_x + k f_y) + \frac{1}{2!} (h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \dots + \frac{1}{n!} (h^n f_{x^n} + \dots + k^n f_{y^n}) + \frac{1}{(n+1)!} (h^{n+1} f_{x^{n+1}} + \dots + k^{n+1} f_{y^{n+1}}) + \dots$

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$\Rightarrow$ 等值线在 (x_0, y_0) 处的切线:

$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0$.

$\Rightarrow$ 考虑在 (x_0, y_0) 外的切面, 有在 P_0 处的所有切线都与正交

则等值面 $f(x, y, z) = C$ 在 (x_0, y_0, z_0) 处的切面:

$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0$

该切面的法线: $\begin{cases} x = x_0 + f_x t \\ y = y_0 + f_y t \\ z = z_0 + f_z t \end{cases}$

Linearization: $L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

误差为 $E(x, y) = f(x, y) - L(x, y)$, 满足 $|E(x, y)| \leq \frac{1}{2} M \cdot ((x - x_0)^2 + (y - y_0)^2)$, 其中 $M = \max\{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$

[Method of Lagrange Multiplier] 拉格朗日乘数法.

$f(x, y, z)$ 在限制域 $g(x, y, z) = 0$ 上的极值点满足 $\nabla f = \lambda \nabla g$.

[Orthogonal Gradient Theorem].

$f(x, y, z)$ 在包含曲线 $C: r(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ 的区域上可微.

设 (x_0, y_0, z_0) 是 f 的极大值/小值点, 则 ∇f 在 (x_0, y_0, z_0) 与 C 正交.

[证]. $\nabla f \cdot r'(t) = (\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}) = \frac{d}{dt} f(x(t), y(t), z(t)) = 0$

两个约束: $\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$. $\nabla f = \lambda \nabla g$ 在 P 点与 C 正交.

[证拉格朗日乘数法].

设 $f(x, y, z)$. $g(x, y, z) = 0$ 可微, P_0 在等值面 $g(x, y, z) = 0$ 上

设 $g \neq 0$ 且 P_0 是 f 的极值点, 则对于等值面上任意穿过 P_0 的可微曲线, f 有极值/大值.

有极值/大值 $\Rightarrow \nabla f$ 与 g 在 P_0 的切向量正交, 而 ∇g 也与 g 在 P_0 处切向量正交 $\Rightarrow \nabla f \parallel \nabla g \Rightarrow \nabla f = \lambda \nabla g$. λ 为拉格朗日乘子.

Vector field.

$F(x, y, z) = M(x, y, z)\vec{i} + N(x, y, z)\vec{j} + P(x, y, z)\vec{k}$. 每点 $\nabla \cdot F = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}$

$\oint_C F \cdot dr = \oint_C M dx + N dy + P dz = \int_a^b F(r(t)) \cdot r'(t) dt = \int_a^b (M \frac{dx}{dt} + N \frac{dy}{dt} + P \frac{dz}{dt}) dt$

沿曲线 C 的线积分.

C 是光滑简单闭曲线. $\oint_C F \cdot dr$ 是 circulation.

光滑闭平面曲线的通量 $= \oint_C M dy - N dx$

守恒 $\Rightarrow \exists \phi, F = \nabla \phi$, ϕ 是 potential function.

$\Rightarrow \oint_C F \cdot dr = \phi(B) - \phi(A)$. 特别的 $\oint_C F \cdot dr = 0$.

[Component Test for conservation fields].

F 是保守场 $\Leftrightarrow F$ 的旋度 $\text{curl}(F) = (\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}, \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}, \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}) = 0$

$\Leftrightarrow \frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} = \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0$.

[Green's Theorem]

定义. 环路密度 circulation density. $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$

通量密度/散度 flux density/divergence $\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$

C 上的正向通量 $\oint_C F \cdot \vec{n} ds = \oint_C M dy - N dx$

曲线表示: 显式 $y = f(x)$ 或 $x = f(y)$. 隐式 $f(x, y) = 0$ 或 $F(x, y) = 0$.

参数: $r(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$. 曲线方程: $r(u, v) = f(u, v)\vec{i} + g(u, v)\vec{j} + h(u, v)\vec{k}$.

$x = f(u, v), y = g(u, v), z = h(u, v)$ (2个参数)

(曲面是 Scalar function)

① Parametrically. $r(u, v) = f(u, v)\vec{i} + g(u, v)\vec{j} + h(u, v)\vec{k}$

面积元 $dA = \sqrt{r_u \times r_v} \cdot \vec{n} dA = \sqrt{f_u^2 + g_u^2 + h_u^2} \sqrt{f_v^2 + g_v^2 + h_v^2} du dv$

② Explicitly. $z = f(x, y)$

$A = \iint_R G(x, y, f(x, y)) \sqrt{f_x^2 + f_y^2 + 1} dx dy = \iint_R G(x, y, z) dA$

③ Implicitly. $F(x, y, z) = C$.

$A = \iint_R \frac{|\nabla F|}{|\nabla F \cdot \vec{n}|} dA$. $\vec{n}$ 是 F 的法向量 $\vec{n} = \nabla F$.

曲面的 (正) 定向 orientation: $\vec{n} = \vec{r}_u \times \vec{r}_v$. 不是所有曲面都有定向性.

对 Vector Field F (线积分 line integral $= \oint_C F \cdot \vec{r}' dt$)

面元的 surface integral $= \text{flux} = \iint_S F \cdot \vec{n} dA = \iint_S F \cdot \vec{n} \sqrt{r_u^2 + r_v^2} du dv$

(向量值函数在空间中运动). Reciprocal Rule $\frac{d}{dt} \left(\frac{1}{f} \right) = -\frac{f'}{f^2}$ (L'Hopital's Rule: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ if $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g'(x) \neq 0$)

Integral Test: $\sum_{n=1}^{\infty} f(n) \approx \int_1^{\infty} f(x) dx$ (for $f(x) > 0$)
2. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ (if $L \neq 0$)
3. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ (if b_n diverges to ∞)

Alternating Series Test: $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ converges if $b_n > 0$, b_n is decreasing, and $\lim_{n \rightarrow \infty} b_n = 0$.
Distance from point to line: $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$
Distance from point to plane: $d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$

Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptical paraboloid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$
Elliptical cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$

Hyperboloid of 1-Sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
Hyperboloid of 2-Sheet: $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
Hyperbolic paraboloid: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}$

Arc Length Formula: $L = \int_a^b |v| dt$
Principal unit normal vector: $N = \frac{1}{|v|} \frac{dv}{ds}$
Binormal vector: $B = T \times N$
Curvature: $\kappa = \frac{|dv/ds|}{|v|^3}$
Torsion: $\tau = -\frac{dB/ds \cdot N}{|v|^3}$

Limit Comparison Test: $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ (if $L > 0$)
2. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ (if b_n diverges to ∞)
3. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ (if b_n diverges to ∞)

Rectifying plane: T, N, B
Normal plane: T, N, B
Tangent and normal scalar components of acceleration: $a_T = \frac{d|v|}{dt}$, $a_N = \frac{|v|^2}{R}$

Fourier Series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L})$
Fundamental period: $T = 2L$
Common period: $T = 2L$

2. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ (if $L \neq 0$)
3. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ (if b_n diverges to ∞)
4. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ (if b_n diverges to ∞)

1. Polar Coordinates: $(r, \theta) \rightarrow (x, y)$
2. Cylindrical Coordinates: $(r, \theta, z) \rightarrow (x, y, z)$
3. Spherical Coordinates: $(\rho, \phi, \theta) \rightarrow (x, y, z)$

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